Alex Tyh Wang

San Francisco, CA | at.alex.wang@gmail.com
linkedin.com/in/alex-tyh-wang | github.com/alextyhwang | alexwang.site

Summary

Computer science undergraduate seeking a software engineering internship/co-op for Spring and/or Summer 2027, with experience in **Python automation**, software testing, debugging, CI/CD, and production systems.

Education

Rochester Institute of Technology — Rochester, NY
Bachelor of Science in Computer Science — Expected May 2028

Skills

Languages and Tools: Python, C, C++, bash scripting, Git, GitHub, Gerrit, Windows, Linux, Makefiles, GDB, IDEs (VS Code)
Systems Fundamentals: Computer architecture, embedded systems, ARM CPU, RISC-V, TrustZone, Caliptra, Trusted Firmware; RTOS architecture, device drivers; SPI, I2C, I3C, UART, USB, PCIe, MDIO
Networking and Security Fundamentals: OSI model, TCP/IP, Ethernet, SERDES, Switches, Routers; IPSec, TLS/SSL, SPDM, digital certificates, cryptography
DSP Fundamentals: Equalizers, Filters, ADC, DAC, FFE, DFE
Software Engineering: Jenkins CI, software regression planning, unit testing, system integration, debugging tools, GUI development; familiarity with lab automation and customer SDKs
Collaboration: Problem-solving, verbal and written communication

Experience

Software Engineer Intern — Jan 2026 – Aug 2026
Paychex — *Rochester, NY*

- Led development of an internal AI developer platform on the **AI Integrations** team, integrating **AI agents and MCP tooling** into developer workflows; presented it to executives.
- Built a custom **software testing and LLM evaluation harness** for developer and Sales workflows to assess reliability.
- Maintained **Jenkins CI/CD pipelines** supporting automated build and deployment workflows; reduced token usage by **55%** with a **subagent architecture** and context engineering.

Student Researcher — Jan 2026 – May 2026
Rochester Institute of Technology — *Rochester, NY*

- Conducted research on large language models and memory.
- Focused on how memory and context shape long-term model behavior.

Software Engineer Intern — Feb 2025 – Aug 2025
Real Wave Capital — *Remote*

- Architected production Angular and TypeScript interfaces for autonomous agents supporting **100+ daily users**; rewrote tool-calling and context strategy to cut token spend by roughly **50%**.
- Integrated APIs into .NET-backed production workflows, improving endpoint response times by **25%** and scaling to **6,000+ weekly requests**.

Projects

iMessages Chatbot Server — Nov 2025 – 2026
Python, asyncio, SQLite, AppleScript — *github.com/alextyhwang/iMessages-Chatbot-Server*

- Built a **Python automation** gateway for macOS Messages with concurrent, stateful conversations, SQLite persistence, and semaphore-controlled API calls for up to **20 concurrent users**.
- Designed asynchronous workflows targeting under **2 seconds** of internal latency and 10–15 sustained messages per minute.

Chatio Agentic Behavior Benchmark — Jun 2025 – Present
Next.js, Python, TypeScript — *chatio.dev*

- Built an LLM benchmark with deterministic Python validation and task-driven tests of real-world usability.
- Engineered a streaming interface and persistence layer using IndexedDB and Cloudflare R2, reducing image storage by **95%**.

Leadership and Activities

Founder and President — Jan 2022 – Present
Bay Area Stem Academy — *San Jose, CA | bayareastemacademy.org*

- Founded a 501(c)(3) nonprofit that taught **660+ students** through **15+ free camps**, partnered with **5+ schools**, and delivered **$35,000+** in free STEM education.

Event Coordinator and Executive Board Member — Sep 2024 – Present
RIT Asian Culture Society — *Rochester, NY*

- Co-lead a community of **200+ members** and organize Night Market, an annual university event with **600 attendees**.